

DPW-8 & AePW-4

All-Hands Meeting and Mini Workshop 3



January 15, 2026

In-Person and Virtual



<https://www.aiaa-dpw.org>

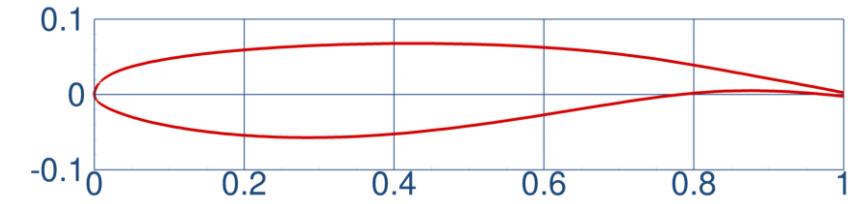
<https://nescacademy.larc.nasa.gov/workshops/AePW4/public>



Test Environment Working Group

Test Case 1a: Workshop-Wide Validation

- **Validation of steady CFD analysis, required**
- **Users are encouraged to employ best practices**
- **Settings**
 - Steady CFD (e.g., RANS)
 - Prefer some version of SA, multiple turbulence models can be submitted
 - Purely 2D simulations (one cell wide)
- **Grids**
 - Six-member RANS grid family; four are required, six are desirable
 - Encourage use of committee-supplied grids; user-generated grids are acceptable
 - Committee-supplied grid is one cell wide with a 230mm chord (same as experiment) and follows RANS best practices
- **Conditions**
 - Mach 0.73, $Re_c=3m$ (based on chord length), $T_{static} = 271 \text{ K (} 487.8 \text{ R)}$
 - Alpha: 1.36, 1.50, 2.50, 3.00, 3.10



ONERA OAT15A Transonic Airfoil

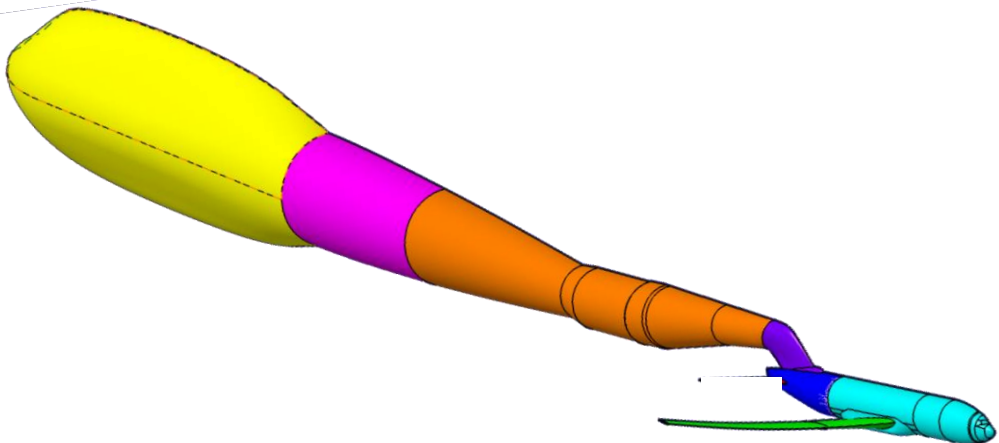
Jaquin, et al. "Experimental Study of Shock Oscillation over a Transonic Supercritical Profiles." AIAA Journal, Vol. 47, No. 9, 2009. Pages 1985-1994.

Test Case 2a: T&I Study (Wing/Body)



- CRM Wing/Body with Upper Swept Strut

1. Wing/Body (2.7% model scale in tunnel)
2. Wing/Body + Upper Swept Strut
 - Wing deformed to matching condition (from DPW7)
 - Aft strut surface (shown in yellow) replaces interface to arc sector



- Geometry

- https://commonresearchmodel.larc.nasa.gov/wp-content/uploads/sites/7/2025/07/DPW-7_WBT_IGES_in_low_q.zip
- https://dpw.larc.nasa.gov/DPW8/Test_Environment/Test_Case_1/Sting_No_Arc_Sector_Bulb_noRotation_2025_09_03.igs

- Conditions & Reference Units

Mach	Re _c	T _{static} (120° F)	α
0.85	5 × 10 ⁶	579.67 R 322.04 K	−1.50°, 0.00°, 1.50°, 2.50°, 2.75°, 3.00°, 3.25, 3.50°, 3.75 4.00°, 4.25°

Sref (semi-span grid)	Cref	Semispan	Moment Center
216.77544 sq.in	7.5195 in	31.23225 in	(156.0003, 0.00, -0.00035)

- Comparison metrics

- Forces / Moments
- Sectional C_p distribution
- Residuals (Flow & Structural Solver)

Comparison Data

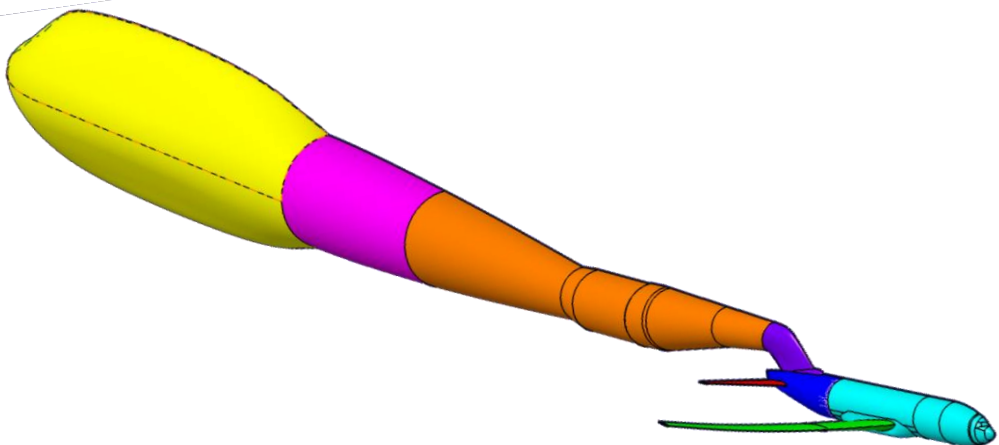
NTF197: r44,r51,r53
NTF215: r43,r103
NTF229: r296,r300,r302
Ames216: r35,r126,r130,r133

Test Case 2b: T&I Study (Wing/Body/Tail)



- CRM Wing/Body with Upper Swept Strut

1. Wing/Body/Horizontal Tail (2.7% model scale in tunnel)
2. Wing/Body/Horizontal Tail + **Upper Swept Strut**
 - Wing deformed to matching condition (from DPW7)
 - Aft strut surface (shown in yellow) replaces interface to arc sector



- Geometry

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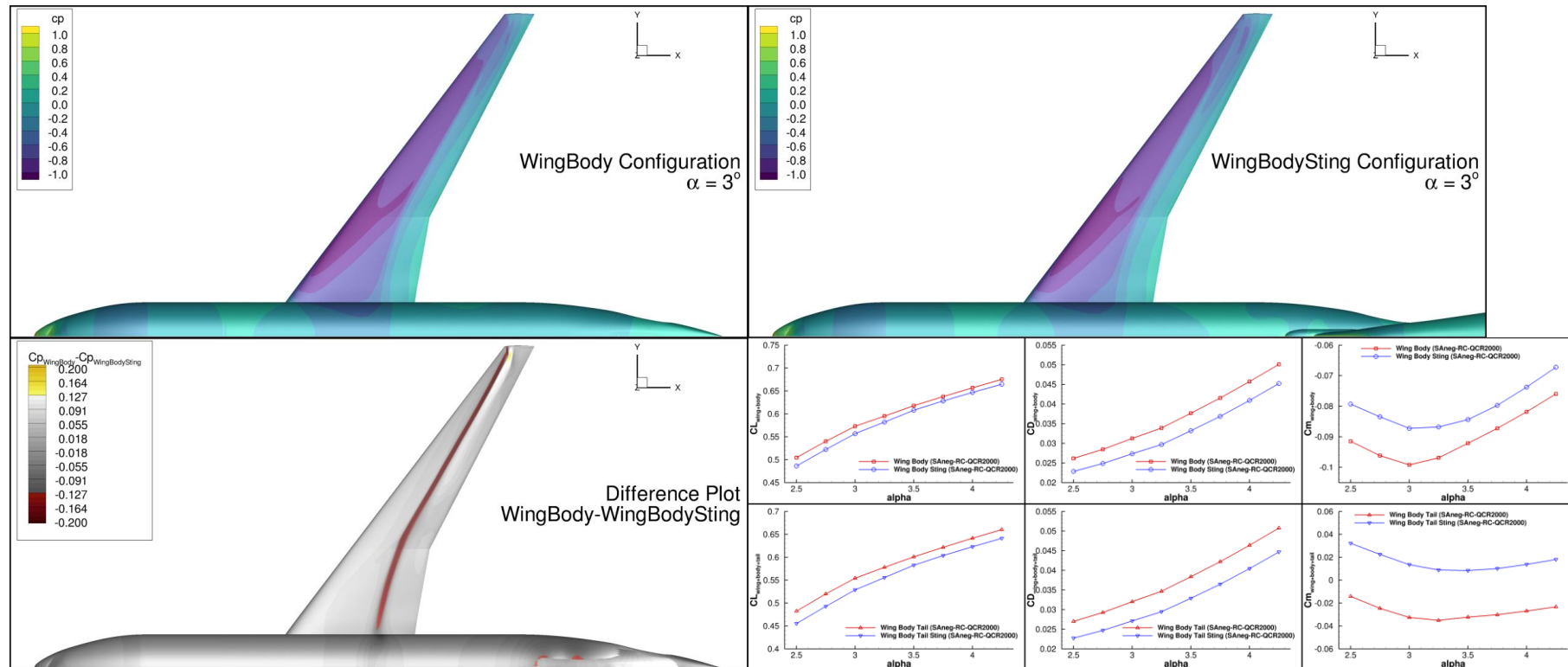
- Comparison metrics

- Forces / Moments
- Sectional C_p distribution
- Residuals (Flow & Structural Solver)

Comparison Data

NTF197: r92,r97,r99
NTF215:
NTF229:
Ames216:

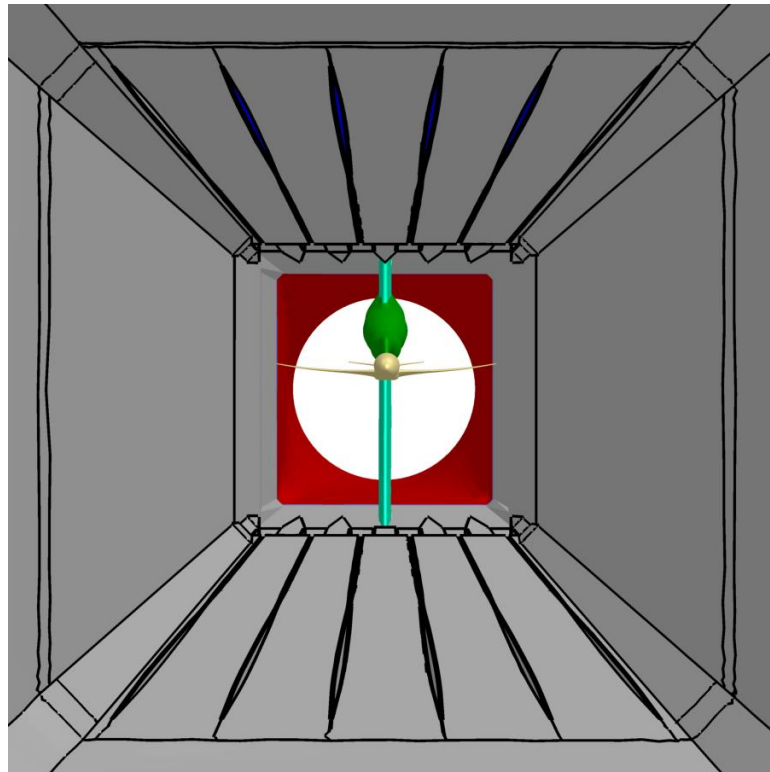
Test Case 2a: Preliminary Findings



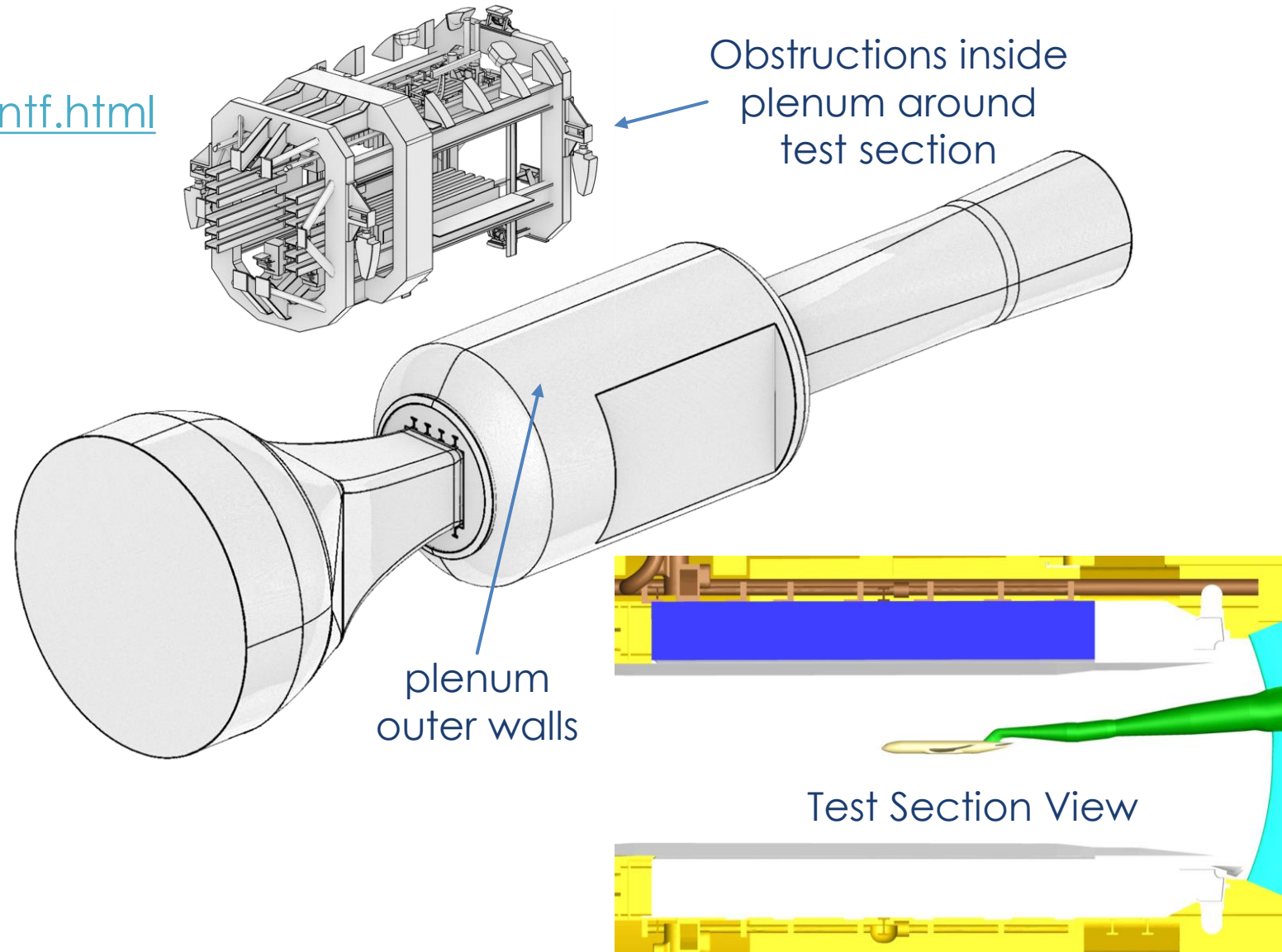
- Preliminary steady-state RANS analysis has been performed using SAneg-RC-QCR2000
- Inclusion of the sting moves the wing shock forward reducing the lift and drag as well as the static stability of the aircraft
- Contour plots at 3-degrees AOA show the shock movement and integrated loads demonstrate the shift caused by the sting

NTF Geometry Available

- **NTF Geometry is available:**
 - <https://www.aiaa-dpw.org/ntf.html>



Test Section View



- How much of the spread between experimental and computational results is due to the test environment?
- What methods are needed to quantify the effect of the mounting hardware on force/moment and pressure measurements?
- Can state-of-the-art methods accurately simulate the full NTF test section, including slots and gaps?

See you in June!

